Does a Lexical Signal Still Help a Visual Retriever?

Reproducing ColPali on ViDoRe DocVQA and testing BM25 fusion

[TODO: First name LAST NAME]

YZM5110 Multimodal Learning — Research Project

[TODO: presentation date]

`github.com/ismailucrn/colpali-hybrid-retrieval`

Motivation: two ways to search PDF pages

Text pipeline

page → OCR / layout → chunks → BM25 or dense embedding

slow ingestion; loses layout, figures; OCR errors

ColPali (Faysse et al.)

page *image* → vision–language model → patch vectors; text query → token vectors; late interaction

no OCR at all

- [Published] ColPali beats every text pipeline on ViDoRe (avg nDCG@5 81.3 vs 67.0).
- [Published] In *text* retrieval, fusing lexical (BM25) and dense rankings is a cheap, well-known gain — but not unconditional (Bruch et al., TOIS 2024).
- [Interpretation] Neural retrievers are often said to struggle with exact strings (numbers, codes, names) — exactly what BM25 matches.

Research question and pre-registered predictions

Question

Does fusing BM25 (over page text) with ColPali improve page retrieval over ColPali alone — and for which query types? Is a gain due to the *lexical* signal, or to ensembling with any second retriever?

Written **before** any fusion run (git commits `162441e`, `55fa350`, `2dfcf31`):

- **P1** C = RRF(BM25, ColPali) > ColPali on test nDCG@5
- **P2** gain concentrated on *exact-match* queries, little on visual ones
- **P3** C > D = RRF(*dense* text retriever, ColPali) (control)

What would weaken it: no gain; gain spread evenly or on visual queries; D $\geq$ C.

How ColPali scores a page

- [Published] PaliGemma-3B encodes the page image; every output vector projected to 128-d.
- [Measured] 1031 vectors per page; 126 MB for 500 pages (bf16).
- Query: tokens + 5 augmentation tokens.
- **Late interaction (MaxSim)**, from ColBERT:

$$s(q,d) = \sum_{i \in q} \max_{j \in d} \langle E_q^{(i)}, E_d^{(j)} \rangle$$

[Interpretation] Each query token finds the patch where its word is *printed* — a kind of "visual lexical matching" without OCR.

We use the authors' checkpoint (v1.3), **inference only**.

Literature: what is known about text + visual fusion

RRF (Cormack 2009)	[Published] beats single runs when fusing *many* runs of similar quality; k=60
Bruch et al. (TOIS 2024)	[Published] default RRF falls *below* the better input on 4/9 datasets
DSE (EMNLP 2024)	[Published] screenshot + text hybrid: "minimal improvement"
IRPAPERS (2026)	[Published] fusion helps (R@1 49 vs 43) — clean LLM transcriptions, text $\approx$ image
Slide study (2025)	[Published] RRF[ColPali, text] 81.0 vs ColPali 82.7 nDCG@10
ViDoRe V1/V2	[Published] visual retrievers evaluated *against* or *without* text, never combined

Gap [Interpretation]

Mixed, uncontrolled results: no significance tests, no "any second retriever" control, no per-query-type analysis, none on a ViDoRe task with the benchmark's OCR.

8 papers in the matrix (`docs/literature_matrix.md`); gap narrowed after a targeted search on 6 Nov — stated in the report.

Reproduction: ColPali on ViDoRe DocVQA

Claim tested: the authors' DocVQA nDCG@5. Tolerance fixed before the run: ± 2 points.

System	Published	source	Measured	Δ
ColPali v1.3	0.585	model card	**0.575**	-0.94 pt
BM25 (OCR text)	0.368	paper Tab. 2	0.391	different OCR
BGE-M3 (OCR text)	0.257	paper Tab. 2	0.249	different OCR

- [Measured] **Reproduced** within tolerance (451 queries, 500 pages, official protocol).
- Authors' code (`colpali-engine`), pinned model + data revisions, Colab L4, bf16.
- Our evaluator matches MTEB exactly (max $|\Delta| = 0$); scores re-evaluated on CPU.
- Deviation: v1.3 instead of the paper's v1.0 (0.544) — v1.0 unsupported in current code.

Experimental design

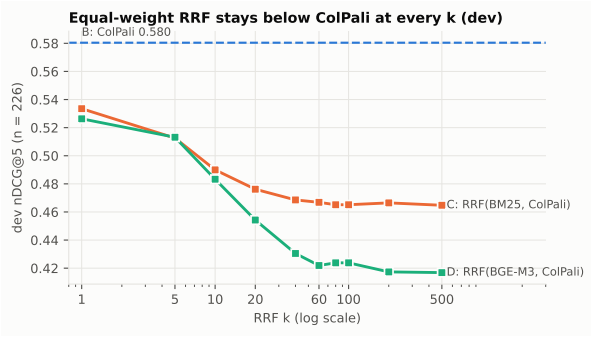
Systems

A	BM25 on Tesseract OCR text	
B	ColPali v1.3	
C	RRF(A, B)	proposed
D	RRF(BGE-M3, B)	control
E	*k* sweep; weighted RRF	ablation

D reads the *same* OCR text, same corpus, same RRF, same tuning budget.

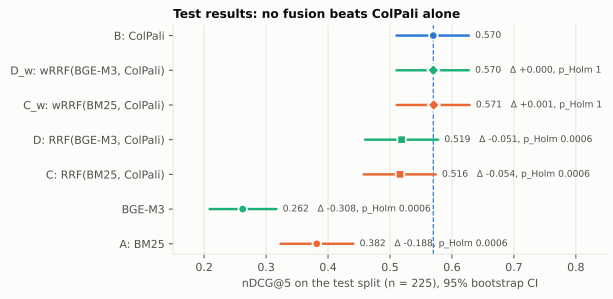
- Query tags from text only, frozen: exact / general / visual
- Split 50/50 stratified: dev 226 / test 225
- *k* and weights chosen on **dev**; test run **once** (script refuses changed settings)
- Bootstrap 95% CIs, paired permutation test, Holm correction
- Resource cost: index time/size, latency, memory

Dev: fusion is below ColPali at every k



- [Measured] ColPali 0.580; best C 0.534, best D 0.526 (both at k=1)
- default k=60: C 0.467
- Weighted RRF (added after this, labelled *dev-informed*): only $w \geq 0.9$ avoids harm; best C_w +0.47 pt on dev
- [Interpretation] the best fusion is the one that changes ColPali least

Test (run once): fusion does not help

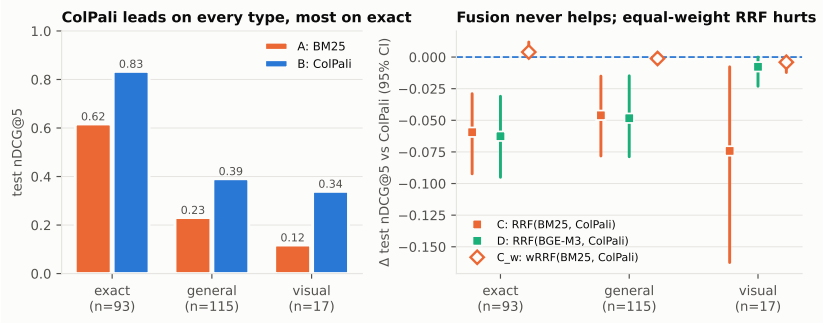


[Measured] nDCG@5, $n = 225$

ColPali **0.570**
C 0.516 Δ $-$**0.054**, p_{Holm} $<$0.001
D 0.519 Δ $-$0.051
C_w 0.571 Δ $+$0.001, p 0.937

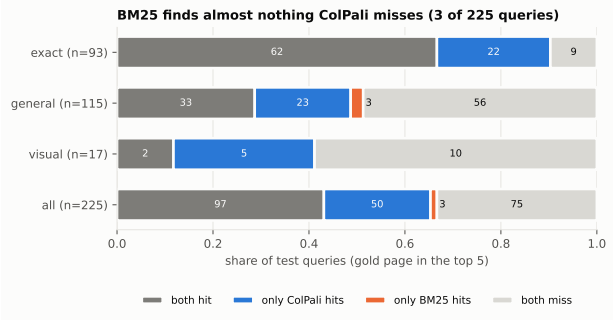
- **P1 rejected** — significant harm
- **P3 not supported** — C$-$D $-$0.003, p 0.844
- dev gain of C_w was selection noise

By query type: the premise of P2 is reversed



[Measured] ColPali is strongest on *exact* queries (0.833 vs BM25 0.617); fusion loses there too (C 0.774). Visual: $n = 17$, wide CIs. **P2 not supported.**

Why: BM25 finds almost nothing ColPali misses

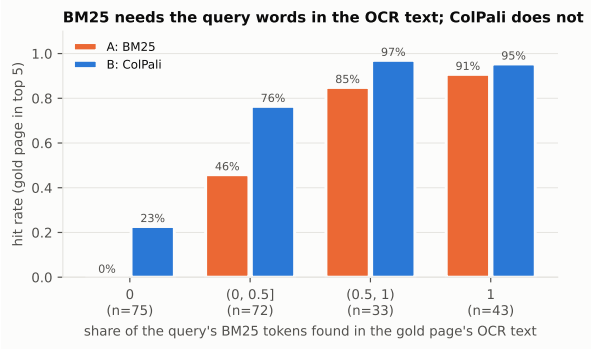


[Measured] gold page in top 5 (test):

- BM25 only: **3** queries (0 exact)
- ColPali only: 50
- per-query oracle: nDCG@5 0.588 vs 0.570 (+1.8 pt)
- C's 49 losses: 42 are queries where BM25 had missed

[Interpretation] little to gain, much to lose — explains P1 and P3

Mechanism: BM25 needs the query's words in the OCR text



[Measured] overlap = share of query tokens found in the gold page's OCR

- zero overlap (75 q.): BM25 0%, ColPali 23%
- full overlap (43 q.): BM25 91%, ColPali 95%

[Interpretation] ColPali reads the page image and still matches when OCR or tokenisation lost the words.

Examples (rank of the gold page)

query	ColPali	BM25	C	why
Unfavourable — C worse than ColPali (49 of 225)				
"What is No. 2-26 of subheading H?"	1	342	3	"No" stopword, "2-26" dropped
"What is the VO. NO.?"	1	137	3	OCR reads "YO. NO."
"... postal stamp at top left?"	3	252	7	page has *empty* OCR
Favourable — C better than ColPali (9 of 225)				
"What is the name of the laboratory?"	14	3	4	real lexical rescue
"How many bullet points are there?"	136	2	4	[Interpretation] probably lucky
"Who is from Indiana?"	2	1	1	typical case: rank 2 → 1

[Measured] all cases: `results/failure_analysis/examples.md`

Post-hoc checks: is it the fusion function, or the tokenizer?

Pre-registered after the main test run (`cb332b7`); chosen on dev, test evaluated once — **exploratory**.

Score-level fusion (Bruch et al.)

$\alpha\,\mathrm{mm}(\text{ColPali}) + (1-\alpha)\,\mathrm{mm}(\text{text})$

- [Measured] dev: every $\alpha < 1$ below ColPali (best 0.575 vs 0.580)
- rule selects $\alpha = 1$ = ColPali alone, for BM25 *and* BGE-M3
- [Interpretation] not specific to rank fusion

Code-preserving BM25 tokenizer

keeps "no" and "2-26"-style codes

- [Measured] BM25 0.382 → 0.405 (Δ +0.023, p 0.027); exact 0.617 → 0.667
- but BM25-only hits stay at 3
- RRF still hurts: Δ −0.045, p_{Holm} <0.001
- [Interpretation] better BM25 only where ColPali is already right

Cost: fusion is cheap — quality is the obstacle

System	Index time (s)	Index size (MB)	Latency seq. (ms)	Latency par. (ms)	Fusion (ms)
A	0.35	0.63	0.12	0.12	0.00
BGE-M3	21.10	1.95	29.04	29.04	0.00
B	111.70	125.85	51.30	51.30	0.00
C	112.04	126.48	51.55	51.43	0.13
D	132.79	127.81	80.47	51.43	0.13
C_w	112.04	126.48	51.56	51.44	0.14
D_w	132.79	127.81	80.47	51.43	0.13

Medians over single queries. Components were measured on different devices: ColPali on an NVIDIA L4 (bf16, peak GPU memory 15.8 GiB), BGE-M3 on Apple M5 MPS (fp16), BM25 and fusion on the M5 CPU. Sequential = sum of components + fusion; parallel = slowest component + fusion.

- [Measured] fusion itself 0.13 ms/query; BM25 index 0.6 MB vs ColPali 126 MB
- parallel retrieval: latency set by ColPali (51.3 ms on L4)
- ColPali peak GPU memory 15.8 GiB [Hypothesis] full-vocabulary logits — not investigated

Limitations

- One task (DocVQA), one OCR (Tesseract), one checkpoint (v1.3)
- [Published] DocVQA is in ColPali's training data (split not stated) — favours ColPali vs zero-shot text baselines
- Test set: 225 queries, only 17 visual — wide CIs, per-type results exploratory
- Score-level fusion and tokenizer tested only post-hoc, one design each (min-max; one tokenizer)
- Main run keeps the official BM25 preprocessing (drops "no", "2-26") for comparability
- Resource numbers from three devices (L4, M5 MPS, M5 CPU)
- Reproduction gap (-0.94 pt) not traced to a cause

Conclusions and open questions

- [Measured] ColPali's DocVQA result reproduces with the authors' code and checkpoint.
- [Measured] Adding BM25 by RRF **hurts** (-0.054 nDCG@5); a dense-text control hurts equally; the best dev-tuned weighting only ties; post-hoc, score-level fusion picks ColPali alone and a better tokenizer adds no headroom.
- [Interpretation] On DocVQA, ColPali already reads the lexical anchors BM25 needs — better than Tesseract OCR does. Fusion with a much weaker ranking imports its errors.
- [Interpretation] Consistent with when fusion fails in text IR (unequal inputs) and when text + visual fusion helped (clean text, equal strength).

Open [Hypothesis]

- Fusion should help once the OCR text itself improves (layout-aware / LLM OCR) or on text-dense corpora — the tokenizer alone was not enough.
- On ViDoRe V2 (less lexical overlap by design), even less.

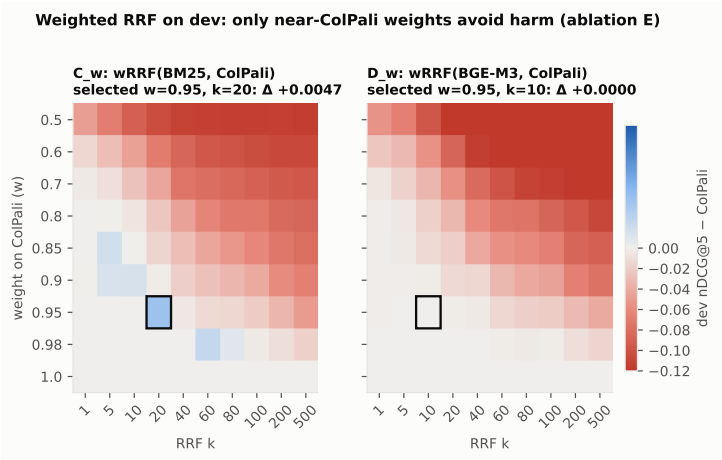
Reproducibility and AI use

- Public repo: configs, pinned dependencies and revisions, committed score matrices, every console log, run registry with failed runs
- Every number in report and slides is generated from `results/` by a script
- [Measured] Fresh-clone check (11 Nov): CPU pipeline re-run without torch — all outputs identical, figures pixel-identical
- Pre-registration is checkable in git history (split, control, ablation committed before the test run)

AI use (disclosed, `docs/ai_usage.md`): Claude Code wrote most code, drove Colab, extracted literature details and drafted documents; the author chose task, checkpoint, control and every pre-registration, approved each run, and verified **[TODO: what you checked]**.

`github.com/ismailucrn/colpali-hybrid-retrieval` tag `v1.0-submission`

Backup: weighted RRF grid on dev



[Measured] ΔnDCG@5 vs ColPali; only $w \geq 0.9$ (ColPali weight) avoids harm.

Backup: full test table

System	Role	nDCG@5 [95% CI]	R@5	R@10	MRR@10	Δ vs B [95% CI]	p_{Holm}
A	text baseline	0.382 [0.323, 0.441]	0.444	0.498	0.369	−0.188 [−0.237, −0.142]	<0.001
BGE-M3	D's text part	0.262 [0.209, 0.317]	0.307	0.418	0.262	−0.308 [−0.364, −0.252]	<0.001
B	reproduced baseline	0.570 [0.510, 0.627]	0.653	0.680	0.546	–	–
C	proposed	0.516 [0.457, 0.574]	0.618	0.667	0.489	−0.054 [−0.076, −0.032]	<0.001
D	control	0.519 [0.460, 0.578]	0.609	0.667	0.496	−0.051 [−0.072, −0.030]	<0.001
C_w	ablation E	0.571 [0.511, 0.629]	0.653	0.680	0.547	+0.001 [−0.002, +0.005]	1.000
D_w	ablation E	0.570 [0.510, 0.627]	0.653	0.680	0.546	+0.000 [+0.000, +0.000]	1.000

Test split, 225 queries, run once with settings frozen on dev. CI: percentile bootstrap over queries (10,000 resamples). p: two-sided paired sign-flip permutation test (10,000), Holm-corrected over the six comparisons with B. [Measured] Post-hoc without the 22 duplicated query texts ($n = 212$): ColPali 0.598, C Δ −0.056, C_w Δ +0.001 — same picture.

Backup: by query type with tests

System	exact (n=93)	general (n=115)	visual (n=17)
A	0.617 (-0.217, $p < 0.001$)	0.231 (-0.160, $p < 0.001$)	0.118 (-0.221, $p = 0.061$)
BGE-M3	0.458 (-0.376, $p < 0.001$)	0.130 (-0.261, $p < 0.001$)	0.088 (-0.251, $p = 0.058$)
B	0.833	0.391	0.339
C	0.774 (-0.059, $p < 0.001$)	0.345 (-0.046, $p = 0.006$)	0.265 (-0.074, $p = 0.128$)
D	0.771 (-0.063, $p < 0.001$)	0.343 (-0.048, $p = 0.002$)	0.331 (-0.008, $p = 1.000$)
C_w	0.837 ($+0.004$, $p = 1.000$)	0.390 (-0.001, $p = 0.743$)	0.335 (-0.004, $p = 1.000$)
D_w	0.833 ($+0.000$, $p = 1.000$)	0.391 ($+0.000$, $p = 1.000$)	0.339 ($+0.000$, $p = 1.000$)

Test nDCG@5 by frozen query tag; in brackets Δ vs B and the uncorrected permutation p (exploratory).